Sign Language System – EDA & Model Evaluation Report

1. Overview

This report summarizes **Exploratory Data Analysis (EDA)** and **Model Evaluation** for the hand landmark sign recognition dataset. The analysis combines:

- **American dataset** (image-based signs)
- **Custom words dataset**
- **Combined landmark dataset** (21-point hand landmarks)
- **KNN classification model**

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2. Dataset Summary

2.1 Source Datasets

Dataset	Description	Link
American	Standard sign dataset	Kaggle
Custom Words	User-defined signs	Manual
Landmarks	Numeric hand vectors	MediaPipe

- **American dataset** includes alphabet letters (A–Z) and numbers (0–9).
- **Custom words** contain static word sign gestures.
- **Landmarks** are numerical hand coordinates extracted using MediaPipe.

2.2 Landmark Data Structure

Attribute	Value
Landmark points per hand	21
Dimensions per point	(x, y, z)
Feature shape	(samples, 21, 3)

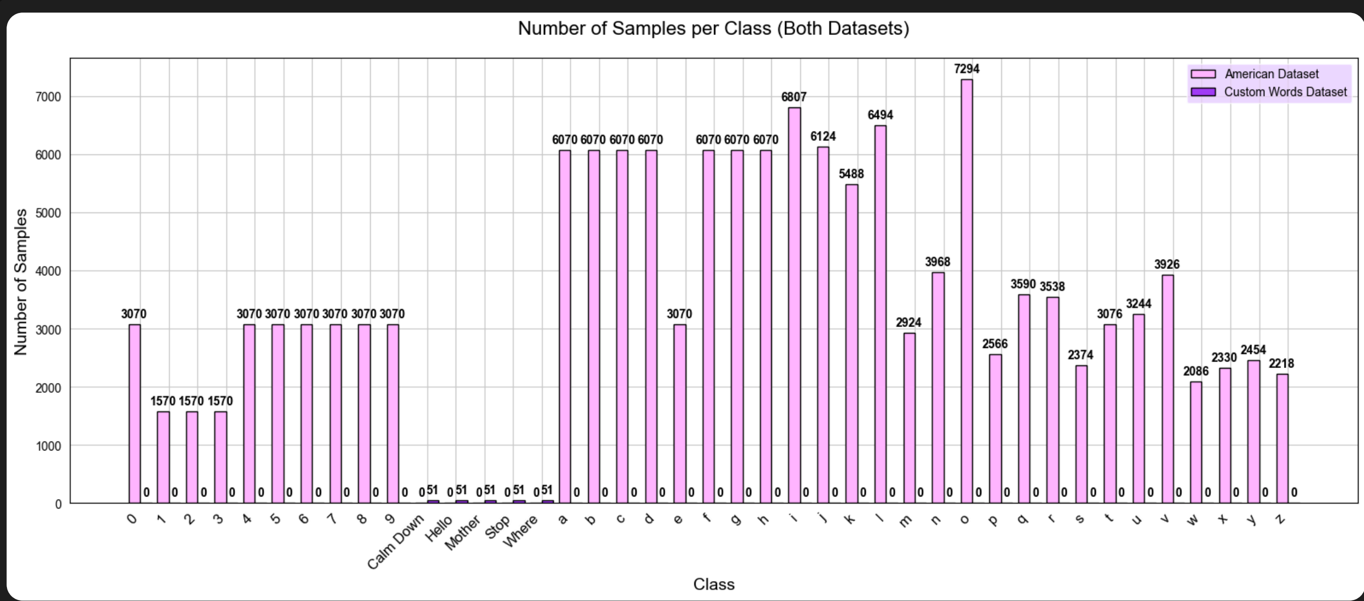
- Each hand consists of **21 landmark points** representing key joints.
- Each point contains **x, y, z** coordinates in 3D space.
- Feature array shape is **(samples, 21, 3)** for model input.

3. Class Distribution Analysis (EDA)

Number of Classes per Dataset

Dataset	Number of Classes
American Dataset	36
Custom Words Dataset	5
Combined Dataset	41

Samples per Class Distribution

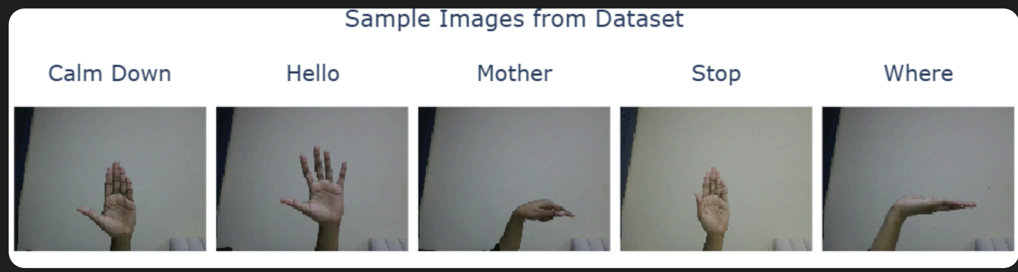


- The dataset exhibits a **strong class imbalance** across gesture categories.
- **Alphabet classes (A–Z)** dominate the dataset, with **thousands of samples per class**.
- **Numeric classes (0–9)** contain significantly fewer samples, mostly under **300**.
- **Custom word classes** are uniformly small, with **50 samples per class**.

Sample Images from Datasets

Ready Dataset

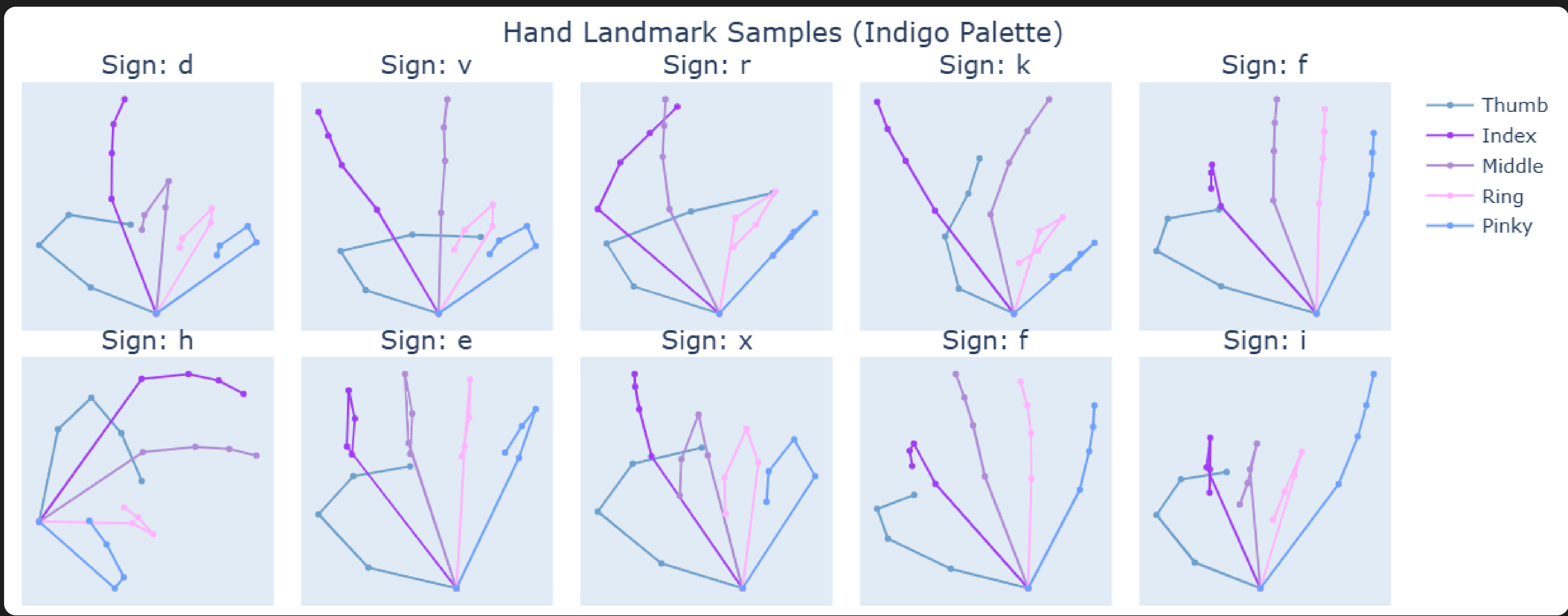
Custom Dataset



- Ready dataset samples originate from a standardized sign language dataset.
- Custom dataset samples were manually collected to represent user-defined gestures.
- Lighting and hand orientation vary across samples.
- Some classes show high visual similarity.

4. Visual Data Exploration

Hand Landmark Samples

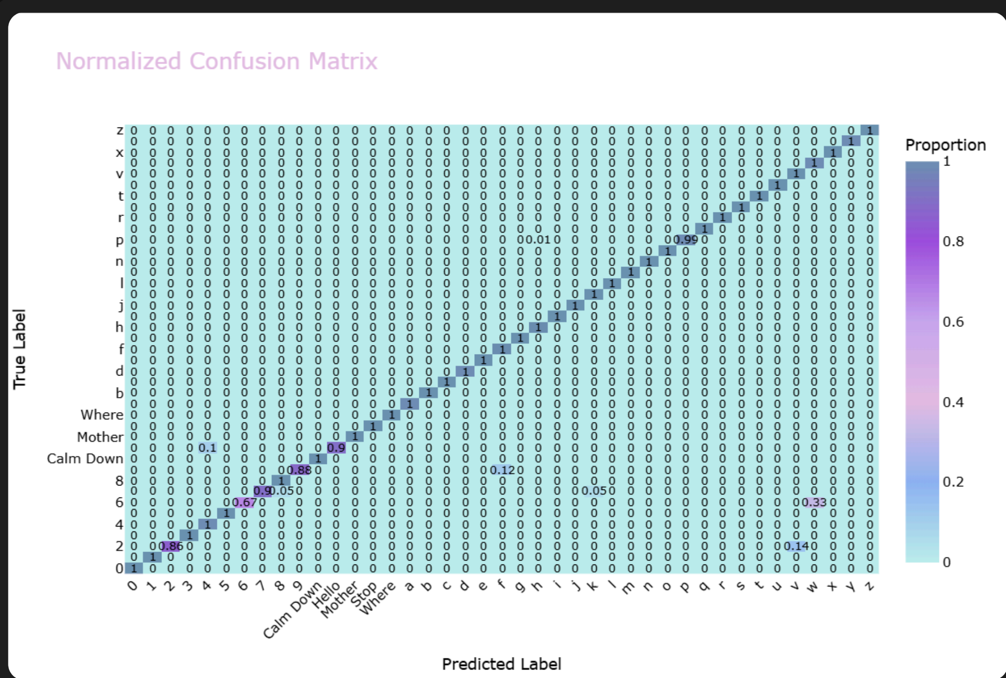


- Visualizes **21 key hand landmarks** per sample using a **5-finger mapping**.
- Helps detect **outliers, inconsistencies, or misaligned landmarks** before model training.

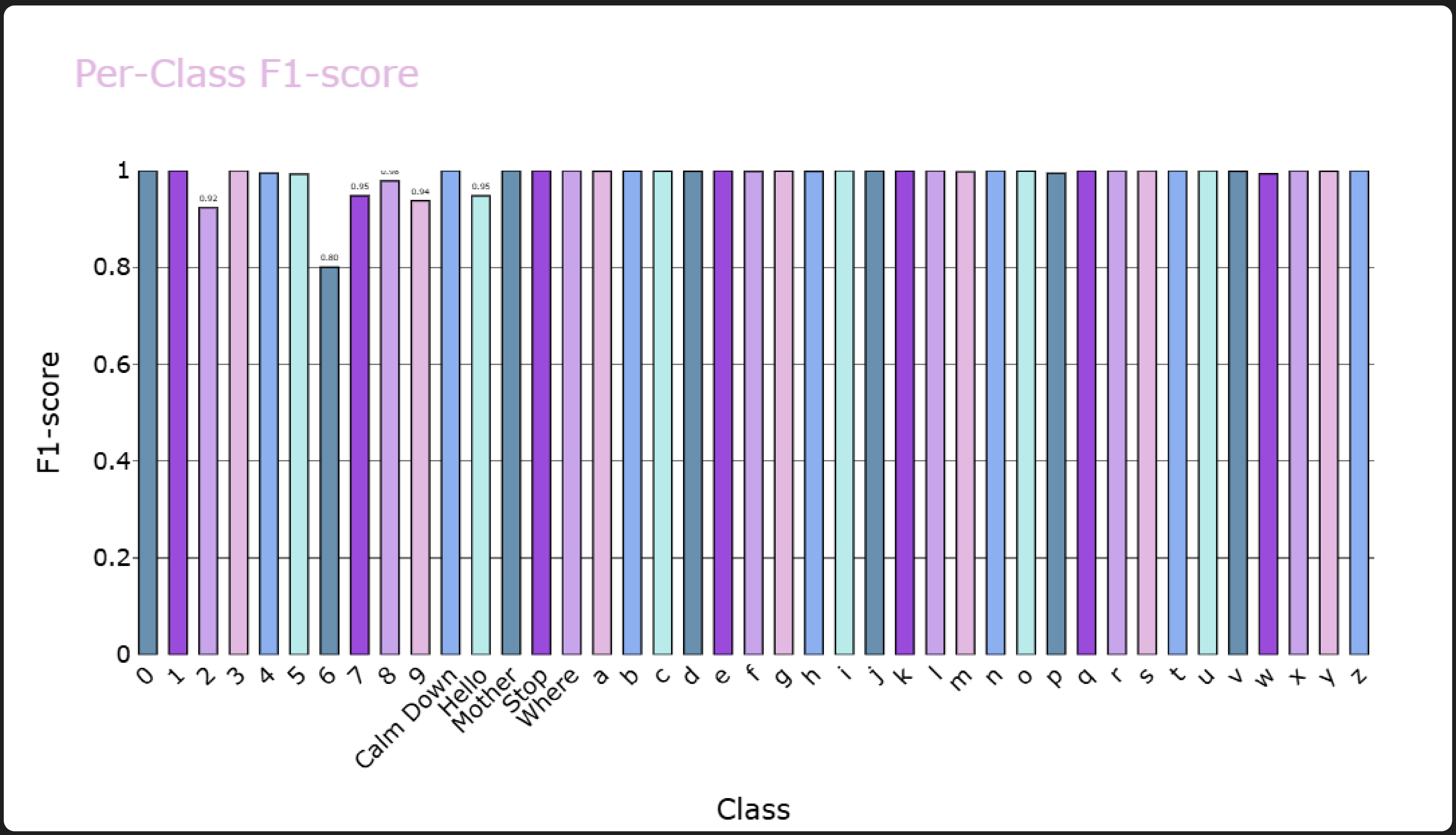
5. Model Evaluation (KNN Classifier)

Normalized Confusion Matrix

- **Overall Accuracy:** Most classes are classified perfectly (1.0) or very close.
- **Minor Misclassifications:**
  - Class 2 → 0.14 misclassified as v
  - Class 6 → 0.33 misclassified as v
  - Class 7 → 0.05 misclassified as 2 and f
  - Class 9 → 0.12 misclassified as f
  - Class Hello → 0.1 misclassified as 4
  - Class p → 0.01 misclassified as h
- **Patterns:** Confusions occur mainly between visually similar hand signs.



Per-Class Performance Metrics



- The model performs exceptionally well, most classes achieve perfect F1-scores.
- A few classes with low samples (2,6,9,Hello,p) show slight misclassification.
- Custom words (Calm Down, Mother, Stop, Where) are recognized accurately (F1 ≥ 0.95).

Global Metrics

- Overall Accuracy: **100%**
- Weighted F1-score: **1.0**
- Macro F1-score: **0.99**

Metric	Score
Macro F1	0.99
Weighted F1	1.0
Accuracy	1.0

Report generated from automated EDA and evaluation pipelines.